

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE	
	Supplementary Examination – Summer 2022	
	Course: B. Tech.	Branch : Computer Science & Engineering
	Semester :VI	
	Subject Code & Name: (BTCOC501), Database Systems	
	Max Marks: 60	Date:
		Duration: 3 Hr.
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.	
		(Level/CO)
		Marks
Q. 1	Solve Any Two of the following.	
A)	Explain purpose of database system.	Remembering
B)	Explain Data Abstraction.	Understanding, applying
C)	Explain data storage & Querying.	Understanding
Q.2	Solve Any Two of the following.	
A)	Give formal definition of tuple relational calculus and explain it.	Remembering
B)	Explain aggregation in relational algebra.	Understanding, applying
C)	Explain domain relational calculus.	Remembering
Q. 3	Solve Any Two of the following.	
A)	Consider following schema & Solve following queries by SQL. Instructor (id, name, department name, salary) Teaches (id, course_id, sec_id, semester, year) Course (couse_id, tittle, department name, credits) Department(department name, building, budget) a. Find the set of all courses taught in summer 2020 semester but not in winter 2021. b. List the names of instructor with salary amounts between 90000 and 100000.	Understanding, analyze
B)	Explain aggregation with example.	Remembering
C)	Explain Function and procedure with example.	Understanding, analyze
Q.4	Solve Any Two of the following.	

A)	Construct B ⁺ tree for the given set of key values (2,3,5,7,11,17,19,23,29,31) using four & six pointers that will fit in one node.	Understading, applying	6
B)	Suppose that we are using extendable hashing on a file that contains records with the following search key values (2,3,5,7,11,19,23,29,31) show the extendable hash structure for this file if the hash function is $h(x) = x \text{ mod } 8$ and buckets can hold three records.	Understanding, applying	6
C)	Explain BCNF.	Understanding, analyze	6
Q. 5	Solve Any Two of the following.		
A)	Explain ACID properties of transaction.	Remembering	6
B)	Explain serializability.	Understanding, analyze	6
C)	Explain deadlock handling.	Understanding,	6
	*** End ***		

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